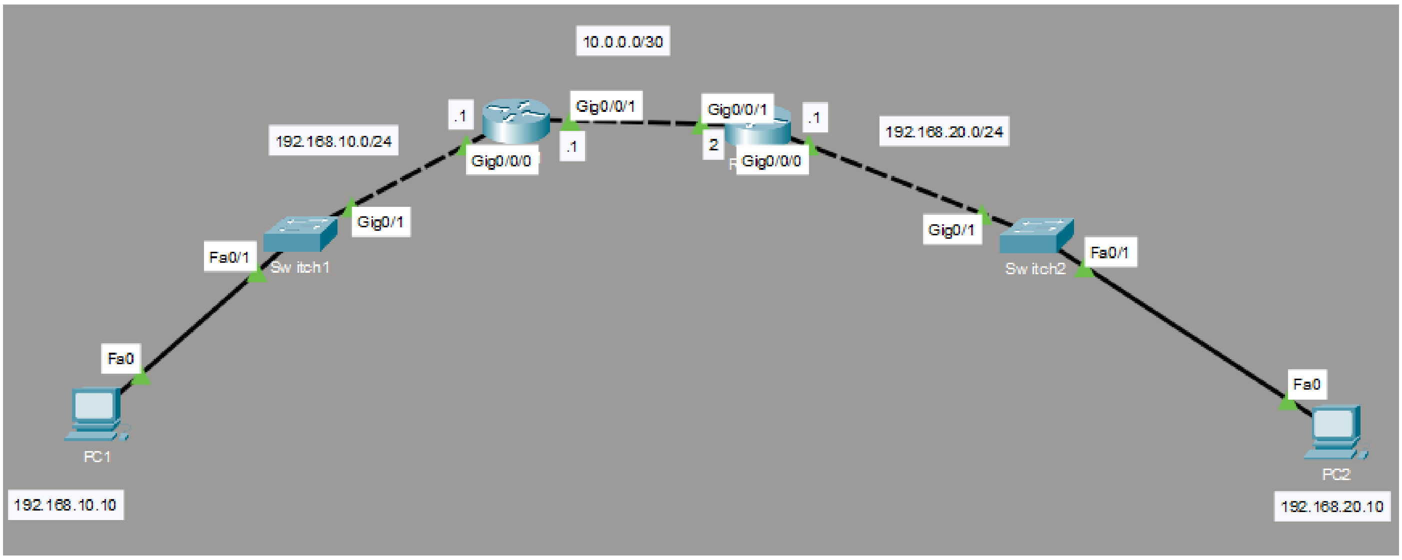
IPv4 & IPv6 Static Routing Lab

Cisco Packet Tracer — Dual-Stack Network Implementation

Project Type: Networking / Routing Lab
Platform: Cisco Packet Tracer
Protocols: IPv4, IPv6, Static Routing, EUI-64
Topology: 2 Routers, 2 Switches, 2 PCs
Status: Completed ✅



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1. Project Overview

This project demonstrates the implementation of a small **dual-stack IPv4/IPv6 network** using Cisco Packet Tracer.

The network consists of two LANs connected through two routers. Each LAN contains a switch and an end device, while the routers provide Layer 3 connectivity between the networks.

The lab was designed to practice:

- IPv4 addressing
- IPv6 addressing
- IPv6 EUI-64
- Static IPv4 routing
- Static IPv6 routing
- VLAN configuration
- Access ports
- Default gateways
- End-to-end connectivity testing
- Basic network troubleshooting

The primary objective was to establish communication between **PC1 and PC2 across both IPv4 and IPv6 networks.**

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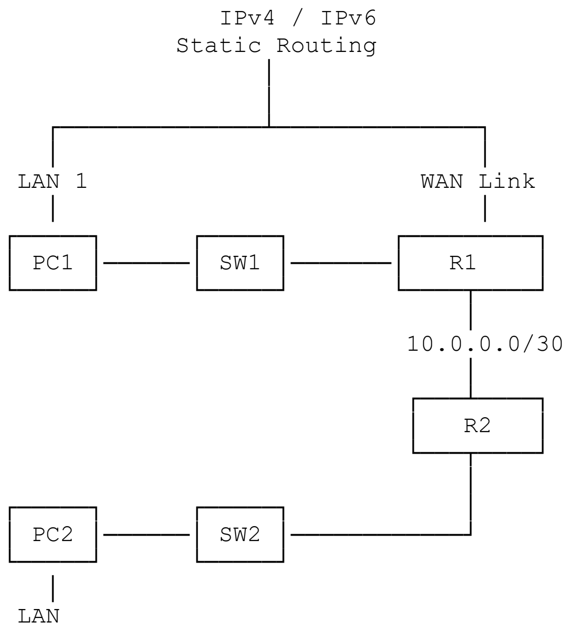
2. Network Topology



Logical design

```
PC1
 │  VLAN 10
SW1
 │
R1
 │  10.0.0.0/30
R2
 │
SW2
 │  VLAN 20
PC2
```

---

3. IPv4 Addressing Plan

Device	Interface	IPv4 Address	Subnet Mask	Purpose
PC1	NIC	192.168.10.10	/24	LAN 1 host
R1	G0/0/0	192.168.10.1	/24	PC1 default gateway
R1	G0/0/1	10.0.0.1	/30	R1-R2 link
R2	G0/0/1	10.0.0.2	/30	R1-R2 link
R2	G0/0/0	192.168.20.1	/24	PC2 default gateway
PC2	NIC	192.168.20.10	/24	LAN 2 host

IPv4 Network Design

```
LAN 1:       192.168.10.0/24
R1:          192.168.10.1

Transit:     10.0.0.0/30
R1:          10.0.0.1
R2:          10.0.0.2

LAN 2:       192.168.20.0/24
R2:          192.168.20.1
```

The /30 network was used for the router-to-router connection because only two usable host addresses are required.

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4. IPv6 Addressing Plan

IPv6 addressing was implemented using **EUI-64** on the router interfaces.

Device	Interface	IPv6 Prefix	Addressing Method
R1	G0/0/0	2001:DB8:10::/64	EUI-64
R1	G0/0/1	2001:DB8:12::/64	EUI-64
R2	G0/0/0	2001:DB8:20::/64	EUI-64
R2	G0/0/1	2001:DB8:12::/64	EUI-64
PC1	NIC	2001:DB8:10::10/64	Manual
PC2	NIC	2001:DB8:20::10/64	Manual

Example R1 EUI-64 addresses

R1 generated:

```
G0/0/0
2001:DB8:10:0:2D0:58FF:FE73:BC01

G0/0/1
2001:DB8:12:0:2D0:58FF:FE73:BC02
```

The interface ID was automatically generated from the router interface's MAC address using the EUI-64 process.

---

5. IPv6 EUI-64 Implementation

IPv6 was enabled globally on both routers:

```
R1(config)# ipv6 unicast-routing
R2(config)# ipv6 unicast-routing
```

The router interfaces were then configured using EUI-64.

R1

```
R1(config)# interface g0/0/0
R1(config-if)# ipv6 address 2001:DB8:10::/64 eui-64
R1(config-if)# no shutdown

R1(config)# interface g0/0/1
R1(config-if)# ipv6 address 2001:DB8:12::/64 eui-64
R1(config-if)# no shutdown
```

R2

```
R2(config)# interface g0/0/0
R2(config-if)# ipv6 address 2001:DB8:20::/64 eui-64
R2(config-if)# no shutdown

R2(config)# interface g0/0/1
R2(config-if)# ipv6 address 2001:DB8:12::/64 eui-64
R2(config-if)# no shutdown
```

Why EUI-64?

EUI-64 allows the router to automatically generate the **64-bit interface ID** from its MAC address.

The `/64` prefix is supplied manually:

```
2001:DB8:10::/64
```

while the remaining 64 bits are generated by the router.

---

6. Switch VLAN Configuration

The two LANs were separated into different VLANs.

Switch	VLAN	Purpose
SW1	VLAN 10	PC1 LAN
SW2	VLAN 20	PC2 LAN

The switch ports connected to the PCs and routers were configured as access ports in their respective VLANs.

SW1

```
SW1(config)# vlan 10
SW1(config-vlan)# name LAN_1

SW1(config)# interface g0/1
SW1(config-if)# switchport mode access
SW1(config-if)# switchport access vlan 10
```

SW2

```
SW2(config)# vlan 20
SW2(config-vlan)# name LAN_2

SW2(config)# interface g0/1
SW2(config-if)# switchport mode access
SW2(config-if)# switchport access vlan 20
```

---

7. PC Configuration

PC1

IPv4

```
IP Address:      192.168.10.10
Subnet Mask:     255.255.255.0
Default Gateway: 192.168.10.1
```

IPv6

```
IPv6 Address:    2001:DB8:10::10/64
IPv6 Gateway:    R1 G0/0/0 EUI-64 address
```

PC2

IPv4

```
IP Address:      192.168.20.10
Subnet Mask:     255.255.255.0
Default Gateway: 192.168.20.1
```

IPv6

```
IPv6 Address:    2001:DB8:20::10/64
IPv6 Gateway:    R2 G0/0/0 EUI-64 address
```

The `::10` interface IDs used on the PCs were manually selected. They are simply valid host/interface identifiers within their respective `/64` networks.

---

8. IPv4 Static Routing

Because R1 and R2 are directly connected only through the `10.0.0.0/30` network, each router requires a route to the remote LAN.

R1

```
R1(config)# ip route 192.168.20.0 255.255.255.0 10.0.0.2
```

This tells R1:

"To reach the `192.168.20.0/24` network, forward traffic to R2 at `10.0.0.2`."

R2

```
R2(config)# ip route 192.168.10.0 255.255.255.0 10.0.0.1
```

This tells R2:

"To reach the `192.168.10.0/24` network, forward traffic to R1 at `10.0.0.1`."

Routing concept

```
PC1
192.168.10.10
       │
       ▼
R1 192.168.10.1
       │
       ▼
10.0.0.2 → R2
       │
       ▼
192.168.20.0/24
       │
       ▼
PC2
192.168.20.10
```

---

9. IPv6 Static Routing

IPv6 static routes were also configured so that each router could reach the remote IPv6 LAN.

The general syntax is:

```
ipv6 route <destination-prefix>/<prefix-length> <next-hop>
```

For example:

```
ipv6 route 2001:DB8:20::/64 <R2-transit-IPv6-address>
```

on R1, and:

```
ipv6 route 2001:DB8:10::/64 <R1-transit-IPv6-address>
```

on R2.

The important distinction when configuring static routes is:

- **Destination network** = where the traffic needs to go
- **Next-hop address** = the neighboring router to which traffic should be forwarded

---

10. Troubleshooting Performed

One of the most useful parts of the lab was troubleshooting an initial connectivity problem.

Problem

PC1 initially could not ping its default gateway:

PC1 → 192.168.10.1

However:

- R1's interface was `up/up`
- R1 had the correct IPv4 address
- R1 had the connected route for `192.168.10.0/24`

This indicated that the problem was likely occurring at **Layer 2**.

Root Cause

The switch interface connected to R1 was not assigned to the same VLAN as PC1.

PC1 was in:

```
VLAN 10
```

while the router-facing switch port was not correctly associated with VLAN 10.

Solution

The router-facing interface on SW1 was configured as an access port in VLAN 10:

```
SW1(config)# interface g0/1
SW1(config-if)# switchport mode access
SW1(config-if)# switchport access vlan 10
```

The same principle was applied to SW2/VLAN 20.

Result

PC1 was then able to successfully reach its default gateway.

Key troubleshooting lesson

A router interface being `up/up` does **not** guarantee that an end device can communicate with it.

The complete path must work:

```
PC
 ↓
Switch Port
 ↓
VLAN
 ↓
Router Interface
 ↓
Layer 3 Routing
```

---

11. Verification Commands

Several Cisco IOS commands were used to verify the implementation.

Verify IPv4 interfaces

```
show ip interface brief
```

Used to verify:

- IP addresses
- Interface status
- Protocol status

Verify IPv4 routing table

```
show ip route
```

Expected entries include:

```
C 192.168.10.0/24
C 10.0.0.0/30
S 192.168.20.0/24
```

on R1.

And:

```
C 192.168.20.0/24
C 10.0.0.0/30
S 192.168.10.0/24
```

on R2.

Verify IPv6 interfaces

```
show ipv6 interface brief
```

This verifies the generated EUI-64 addresses.

Verify IPv6 routing table

```
show ipv6 route
```

The routing table should contain:

```
C  Connected
L  Local
S  Static
```

for the appropriate IPv6 networks.

---

12. Connectivity Testing

Connectivity was tested progressively rather than immediately testing PC1 → PC2.

IPv4

```
PC1 → R1
       ↓
R1 → R2
       ↓
R2 → PC2
       ↓
PC1 → PC2
```

IPv6

```
PC1
 ↓
R1 LAN IPv6
 ↓
R1-R2 IPv6 link
 ↓
R2 LAN IPv6
 ↓
PC2
```

Final Result

✅ PC1 successfully communicated with PC2 using IPv6.

This confirmed that:

- IPv6 addressing was functioning
- EUI-64 addressing was functioning on the router interfaces
- IPv6 forwarding was enabled
- The routers had working paths between the LANs
- PC default gateways were correctly configured
- End-to-end IPv6 connectivity was established

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13. Key Concepts Demonstrated

Layer 2 — Switching

- VLAN creation
- Access ports
- VLAN-to-port assignment
- Understanding the relationship between end devices and switch VLANs

Layer 3 — IPv4

- IPv4 addressing
- Default gateways
- /30 point-to-point networks
- Static routes
- Next-hop addressing
- Routing table verification

Layer 3 — IPv6

- 128-bit IPv6 addressing
- /64 prefixes
- EUI-64
- IPv6 forwarding
- IPv6 static routing
- IPv6 routing-table verification

Troubleshooting

- Layer 2 vs. Layer 3 troubleshooting
- Interface status verification
- VLAN mismatch diagnosis
- Route verification
- End-to-end connectivity testing

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14. Skills Demonstrated

Networking

IPv4 · IPv6 · Subnetting · VLANs · Static Routing · EUI-64 · Default Gateways · Layer 2/Layer 3 Troubleshooting

Cisco

Cisco IOS · Packet Tracer · Interface Configuration · Routing Tables · IPv6 Configuration

Troubleshooting Methodology

Physical → Layer 2 → Layer 3 → Routing → End-to-End Verification

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15. Project Outcome

The completed lab successfully implemented a functional **dual-stack network** connecting two separate LANs through two Cisco routers.

The project provided hands-on experience with both IPv4 and IPv6 addressing and reinforced the relationship between **switching, VLANs, routing, and end-device configuration.**

A particularly important takeaway was understanding that successful routing depends on the **entire communication path**, not simply the presence of an IP address or routing table entry.

Portfolio Summary

Dual-Stack IPv4/IPv6 Static Routing Lab — Cisco Packet Tracer

Designed and implemented a two-router, two-switch enterprise-style network supporting IPv4 and IPv6 connectivity. Configured VLANs, static IPv4/IPv6 addressing, IPv6 EUI-64, default gateways, and static routing. Troubleshot a Layer 2 VLAN mismatch that prevented initial gateway connectivity and verified end-to-end communication between hosts using IPv4 and IPv6.

Technologies

Cisco Packet Tracer · Cisco IOS · IPv4 · IPv6 · VLAN · Static Routing · EUI-64 · TCP/IP · Network Troubleshooting

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Verification screenshots

ip interface brief

R1/R2

```
Router1#SH IP INTerface BR
Router1#SH IP INTerface BRief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   192.168.10.1    YES manual up                    up
GigabitEthernet0/0/1   10.0.0.1        YES manual up                    up
GigabitEthernet0/0/2   unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router1#
```

```
Router2#show ip inter
Router2#show ip interface br
Router2#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   192.168.20.1    YES manual up                    up
GigabitEthernet0/0/1   10.0.0.2        YES manual up                    up
GigabitEthernet0/0/2   unassigned      YES unset  down                  down
Vlan1                  unassigned      YES unset  up                    down
Router2#
```

ip route

R1/R2

```
Router1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/0/1
L       10.0.0.1/32 is directly connected, GigabitEthernet0/0/1
     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0/0
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0/0
S    192.168.20.0/24 [1/0] via 10.0.0.2

Router1#
```

```
Router2#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/0/1
L       10.0.0.2/32 is directly connected, GigabitEthernet0/0/1
S    192.168.10.0/24 [1/0] via 10.0.0.1
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/0/0
L       192.168.20.1/32 is directly connected, GigabitEthernet0/0/0

Router2#
```

show ipv6 interface brief — showing your EUI-64 addresses

R1

```
Router1#sh ipv6 interface br
Router1#sh ipv6 interface brief
GigabitEthernet0/0/0        [up/up]
    FE80::2D0:58FF:FE73:BC01
    2001:DB8:10:0:2D0:58FF:FE73:BC01
GigabitEthernet0/0/1        [up/up]
    FE80::2D0:58FF:FE73:BC02
    2001:DB8:12:0:2D0:58FF:FE73:BC02
GigabitEthernet0/0/2        [administratively down/down]
    unassigned
Vlan1                       [administratively down/down]
    unassigned
Router1#
```

R2:

```
Router2#sh ipv6 interface br
Router2#sh ipv6 interface brief
GigabitEthernet0/0/0        [up/up]
    FE80::2E0:F7FF:FE02:1A01
    2001:DB8:20:0:2E0:F7FF:FE02:1A01
GigabitEthernet0/0/1        [up/up]
    FE80::2E0:F7FF:FE02:1A02
    2001:DB8:12:0:2E0:F7FF:FE02:1A02
GigabitEthernet0/0/2        [up/down]
    unassigned
Vlan1                       [up/down]
    unassigned
Router2#
```

show ipv6 route

R1:

```
Router1#sh ipv6 route
IPv6 Routing Table - 6 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       D - EIGRP, EX - EIGRP external
C   2001:DB8:10::/64 [0/0]
     via GigabitEthernet0/0/0, directly connected
L   2001:DB8:10:0:2D0:58FF:FE73:BC01/128 [0/0]
     via GigabitEthernet0/0/0, receive
C   2001:DB8:12::/64 [0/0]
     via GigabitEthernet0/0/1, directly connected
L   2001:DB8:12:0:2D0:58FF:FE73:BC02/128 [0/0]
     via GigabitEthernet0/0/1, receive
S   2001:DB8:20::/64 [1/0]
     via 2001:DB8:12:0:2E0:F7FF:FE02:1A02
L   FF00::/8 [0/0]
     via Null0, receive
Router1#
```

R2:

```
Router2#sh ipv6 route
IPv6 Routing Table - 6 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       D - EIGRP, EX - EIGRP external
S   2001:DB8:10::/64 [1/0]
     via 2001:DB8:12:0:2D0:58FF:FE73:BC02
C   2001:DB8:12::/64 [0/0]
     via GigabitEthernet0/0/1, directly connected
L   2001:DB8:12:0:2E0:F7FF:FE02:1A02/128 [0/0]
     via GigabitEthernet0/0/1, receive
C   2001:DB8:20::/64 [0/0]
     via GigabitEthernet0/0/0, directly connected
L   2001:DB8:20:0:2E0:F7FF:FE02:1A01/128 [0/0]
     via GigabitEthernet0/0/0, receive
L   FF00::/8 [0/0]
     via Null0, receive
Router2#
```

PC2 successful IPv4/IPv6 ping to PC1

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 2001:db8:10::10

Pinging 2001:db8:10::10 with 32 bytes of data:

Reply from 2001:DB8:10::10: bytes=32 time<1ms TTL=126
Reply from 2001:DB8:10::10: bytes=32 time<1ms TTL=126
Reply from 2001:DB8:10::10: bytes=32 time<1ms TTL=126
Reply from 2001:DB8:10::10: bytes=32 time<1ms TTL=126

Ping statistics for 2001:DB8:10::10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 192.168.10.10: bytes=32 time<1ms TTL=126
Reply from 192.168.10.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

The VLAN configuration on SW1/SW2

```
Switch1#sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24, Gig0/2
10   IT                               active    Fa0/1, Gig0/1
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch1#
```

```
Switch2#sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24, Gig0/2
20   NOC                              active    Fa0/1, Gig0/1
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch2#
```